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***Real-Time Motor Fatigue Detection using
Hardware-Accelerated Signal Processing
and ML on PYNQ Z2***

(In partial fulfillment of the requirement for the B.E.E.T.C.E.)

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ABSTRACT

Induction motors, widely used across various industries, are prone to insulation fatigue due to factors like overvoltage, overheating, excessive vibrations, harsh environmental conditions, and inadequate maintenance. This fatigue can result in multiple motor faults, posing serious risks to industrial operations and potentially causing complete system shutdowns. Therefore, early fault detection is crucial to prevent severe disruptions. Traditional methods such as thermal imaging, oil analysis, and manual monitoring are commonly used but fall short in providing real-time fault detection. While advanced techniques like remote condition monitoring offer improved efficiency, they often require significant resources, making them less practical for widespread implementation.

This research addresses the shortcomings of conventional fault detection methods by introducing an innovative approach based on Park's Vector Analysis. Unlike offline techniques, our system operates in near real-time, capturing winding current data using a Data Acquisition System (DAS) and analyzing it simultaneously. This significantly enhances the speed and accuracy of fault detection, thereby improving industrial reliability and safety.

Furthermore, the integration of SoC technology allows seamless scalability, making it adaptable for various industrial applications, from small-scale manufacturing units to large power plants. By enabling real-time data processing and fault diagnosis, this system minimizes downtime and maintenance costs, ultimately enhancing operational efficiency. As industries continue to adopt automation and smart monitoring systems, the proposed solution paves the way for more reliable and intelligent motor health management, ensuring sustained productivity and safety in industrial environments.

Additionally, the proposed fault detection system is implemented on a System on Chip (SoC) platform, leveraging a hybrid architecture that integrates a Processing System (PS), akin to a Central Processing Unit (CPU), with Programmable Logic (PL) for data analysis. This hardware-software co-design offers greater system flexibility while substantially reducing detection latency compared to traditional software-based solutions.

INTRODUCTION

Induction-based machines are vital to modern industries, ensuring the seamless operation of essential processes across sectors such as production and transportation. Key components like induction motors and electrical transformers serve as the backbone of industrial infrastructure. Their reliability is crucial, as any failure can lead to unplanned downtime, production losses, and significant repair expenses. Beyond financial implications, such disruptions can also impact the livelihoods of workers and the communities that rely on these industries.

Given their critical role, it is essential to identify the root causes of faults in these machines and implement effective strategies to maintain their health and prevent failures. This study aims to analyse the causes of faults in induction motors and develop an efficient system capable of detecting or predicting potential failures.

1. Induction Motor Fatigue

An induction motor is an AC-powered electric motor consisting of a rotor, stator, and coils that convert electrical energy into mechanical energy through electromagnetic induction. Also known as asynchronous motors, they are the most widely used motors across industrial, commercial, and residential applications. Induction motors operate on the principle of electromagnetic induction. When the rotating magnetic field of the stator interacts with the stationary rotor, it induces an electromagnetic field in the rotor. The electric current generated in the rotor, responsible for producing torque, is obtained through electromagnetic induction from the stator winding's magnetic field.

This study focuses on a three-phase induction motor, which consists of a stator and a rotor. The stator is the stationary component, featuring a steel frame that encloses a hollow cylindrical core made of thin silicon steel laminations to minimize eddy current and hysteresis losses. The stator contains a three-phase winding known as the stator winding. The rotor, in turn, carries a short-circuited winding called the rotor winding. When a three-phase current is supplied, it generates an electromagnetic field in the stator winding, which induces torque in the rotor winding, driving the motor.

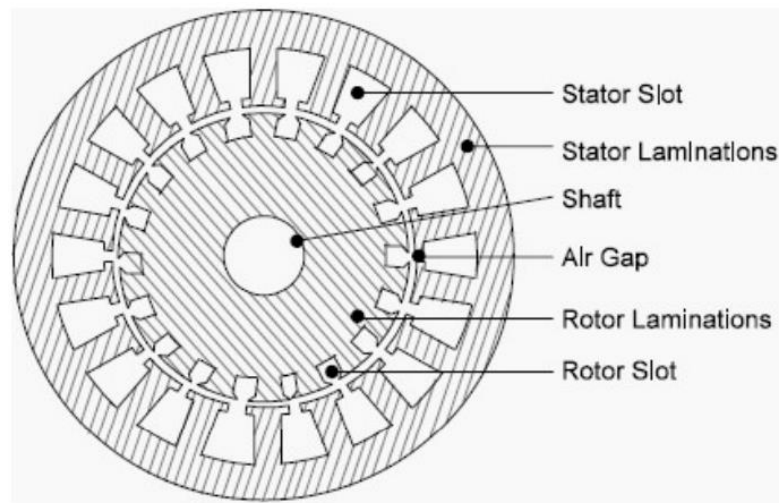


Fig.1.: Basic construction of a Three-phase motor[1]

Electrical fatigue refers to the gradual degradation of motor components due to repeated electrical stress. This irreversible process occurs when a crystalline material undergoes cyclic loading over time. In induction motors, electrical fatigue primarily affects the insulation system, resulting from factors such as voltage spikes, harmonics, and frequent starts and stops common challenges in industrial environments.

Several factors contribute to the fatigue of these motors:

1. **Voltage Fluctuations:**

Sudden spikes or drops in supply voltage can compromise the insulation system, these sudden fluctuations contribute in accelerating wear and tear over time.

2. **Harmonics:**

Nonlinear loads in the electrical system introduce harmonics to the system, causing additional stress on motor windings and insulation, that may cause wearing in the wires.

3. **Frequent Starts and Stops:**

The thermal cycling resulting from frequent starts and stops induces expansion and contraction, leading to insulation degradation of the wires used in windings.

4. **Overloading:**

Operating motors beyond their rated capacity generates excessive heat, hastening insulation breakdown and overall wear of the system.

5. **Environmental Conditions:**

Harsh environmental factors, such as high temperatures or excessive humidity, can

contribute to the deterioration of insulation materials and introduce fatigue conditions in the machines.

6. Inadequate Cooling:

Poor ventilation or poor cooling systems can cause overheating or thermal energy storage in the system that adversely affects the motor's insulation and longevity by putting off the thermodynamical balance of the system.

7. Contaminants:

The presence of dust, dirt, or corrosive substances in the motor's vicinity can compromise insulation integrity. Sometimes poor materials used in the motor windings compromise the system's health.

8. Mechanical Vibration:

Excessive mechanical vibrations can weaken insulation and contribute to the degradation of winding materials. Also, excessive vibrations may cause structural faults that can cause total breakdown.

9. Age and War:

As motors age, the insulation system naturally degrades, making them more susceptible to electrical fatigue.

10. Inadequate Maintenance:

Lack of regular maintenance, including cleaning, lubrication, and inspection, can allow issues to accumulate and escalate.

2. Health monitoring methods

For efficient operation of any industry maintaining the health of induction motor is very important and over the time several methods[2] of monitoring induction motor's condition are devised. Some of them are:

1. Manual monitoring:

One of the oldest methods of condition monitoring is manual condition monitoring, which is done by periodical checking of the target motor by human observation.

2. Oil testing:

Oil testing is another method for monitoring an induction machine; this method is generally implemented to condition monitor the transformers. In this method the insulating oil used in the induction machine is tested for contamination.

3. Insulation Resistance measurement:

Oil testing is another method for monitoring an induction machine; this method is generally implemented to condition monitor the transformers. In this method the insulating oil used in the induction machine is tested for contamination.

4. Thermal Imaging:

Infrared cameras are employed to take thermal images of the target motor and those images are analysed, currently the analysis is done using machine learning. Hotspots signify potential issues such as winding faults or inadequate cooling in the target motor.

5. Acoustic Monitoring:

The noise made by the motor is observed, any abnormality in the noise denotes possible fault in target motor.

6. Power Quality Analysis:

In power quality analysis, the quality of the electrical power supply is examined. In this method harmonics, voltage fluctuations, and other irregularities impacting motor health are detecting.

7. Current signature analysis:

The electrical current waveform is noted while the motor is in action and it is analysed for finding any abnormal pattern. Any abnormal pattern denotes fault or possibility of fault in the motor.

8. Vibration Analysis:

The vibration generated while the motor is in action is measured using vibration sensors. Abnormal vibrations can indicate misalignments, mechanical imbalances, or bearing defects.

9. IoT and Remote Condition Monitor:

It is a more modern approach towards condition monitoring, where the concept of Internet of Things and Remote Monitoring are applied. Different sensors are used to collect data as required, for Remote monitoring the data is collected remotely and analysed to detect fault or possibility of future fault.

3. SoC

Traditional condition monitoring methods are often labour-intensive and resource-demanding. However, advancements in technology provide a more efficient alternative, with System-on-Chip (SoC) emerging as a promising solution. An SoC is a compact, integrated circuit that combines all essential components of a computer system onto a single chip.

SoCs offer multiple functionalities a smaller form factor while consuming less power, making them highly effective for various industrial applications. Additionally, they incorporate storage capacity and utilize Field Programmable Gate Arrays (FPGAs), which enable significantly faster data processing compared to conventional CPU-based systems. The FPGA's inherent parallelism and on-field programmability allow real-time data analysis with minimal latency. The key improvements that SoC will bring to the table are:

- 1. Real-time Data Processing:**

SoC's inherent processing capabilities enable real-time data analysis from various sensors monitoring the motor.

- 2. Sensor Integration:**

SoC can seamlessly integrate with a variety of sensors, including accelerometers for vibration monitoring, temperature sensors for thermal analysis, and current sensors for electrical parameter evaluation.

- 3. Machine Learning Integration:**

SoC can host machine learning algorithms directly on the chip. This allows for on-device analysis of historical data, enabling the system to learn and adapt to evolving motor conditions over time.

- 4. Communication Capabilities:**

Industrial Internet of Things (IIoT) platforms. This connectivity ensures that the condition data can be relayed in real-time to central monitoring systems for further analysis and decision-making.

- 5. Compact Form Factor:**

The compact nature of SoC makes it suitable for embedded applications within motor control units. Its small size allows for easy integration into existing motor control systems without significant hardware modifications.

- 6. Energy Efficiency:**

SoC's energy-efficient design aligns with the demand for sustainable industrial

practices. It allows for continuous monitoring without imposing a substantial energy burden on the overall system.

. Utilizing SoC for condition monitoring enables industries to shift from reactive to proactive maintenance. Real-time insights help detect issues early, reducing downtime, lowering costs, and enhancing the efficiency of induction motors.

LITERATURE REVIEW

As electrical systems grow increasingly intricate and interconnected, the demand for efficient and prompt maintenance of electrical machinery becomes ever more essential. The complex interactions between components and systems in contemporary electrical machinery necessitate careful attention and proactive maintenance to ensure optimal performance and durability. Inadequate maintenance can lead to expensive downtime, diminished efficiency, and potentially dangerous malfunctions. Therefore, establishing comprehensive maintenance protocols is vital to maintain the reliability and functionality of electrical systems in our rapidly advancing technological environment. The previous chapter established that induction motors are indispensable in modern industry and daily life, yet susceptible to various faults that may eventually cause motor breakdown and subsequent system failure.

2.1 Health Monitoring System for Induction Motor

According to this paper[3], research has demonstrated that broken bar fault represents a significant cause of failure in induction motors. The researchers simulated a 5 hp programmable induction motor and evaluated it under various healthy and faulty conditions to collect data for fault diagnosis, revealing broken bars as a primary source of malfunction. The findings of paper[4] indicated that static air-gap eccentricity can induce faults in induction motors; the researchers thoroughly examined this fault and sought an alternative detection method for squirrel cage motors. They introduced an approach based on analyzing line neutral voltage occurring between the supply and stator neutrals to detect air-gap static eccentricity, and emphasized the classification of all RSH harmonics in line neutral voltage. Paper[5] outlined an off-line power transformer diagnosis utilizing dynamic resistance measurement. Multiple diagnostic measurements were conducted using a single device. Examples and case studies were discussed to demonstrate the variety of defects and degradation mechanisms detectable through this procedure. Notably, the most common OLTC defects proved detectable. The measurements exhibited high sensitivity to degradation from long-term aging and OLTC maintenance errors. A substantial population of OLTCs underwent testing, with many showing contact degradation. Finally, several areas requiring careful consideration were addressed, including test current amplitude, circuit resistance, secondary short circuiting, and winding configuration. The research in paper[6] proposed an off-line power transformer diagnosis. Multiple diagnostic measurements were achievable using a single device. Examples and case studies illustrated the range of defects and degradation mechanisms detectable through this procedure. In particular, the most common OLTC defects could be identified. The measurements showed significant sensitivity to degradation from long-term aging and OLTC maintenance errors. A large OLTC sample was tested, with numerous units exhibiting contact degradation. Finally, several areas requiring careful attention were discussed, including test current amplitude, circuit resistance, secondary short circuiting, and winding configuration. The system developed in paper[7] demonstrated a more traditional approach for monitoring induction motors or generators through partial discharge testing. Partial discharges (corona) indicate most types of deterioration in motor and generator stator windings rated 4 kV and above. Experience shows that partial discharges precede failure by years, providing adequate time to schedule corrective maintenance and prevent in-service machine failure. This paper described a novel partial discharge test performable by plant personnel during normal machine operation, enabling all users of large machines to plan maintenance. The key requirement of this new

test involves eliminating interference from other sparking and discharging sources in a plant, which previously caused false indications of stator winding deterioration. The new test employs high-voltage capacitive sensors, high-frequency current transformers, or 1000 MHz electromagnetic couplers to detect discharge signals. These signals undergo pulse-by-pulse processing by specialized electronic instruments to eliminate noise, with the remaining partial discharges categorized by number, magnitude, and phase position. The test has been implemented on over 100 machines with promising results. Paper[8] examined in-turn short circuits in stator windings. Early detection of interturn shorts during motor operation would prevent consequential damage to adjacent coils and the stator core, reducing repair costs and motor outage time. Beyond the benefits of early turn insulation breakdown detection, significant advantages would result from locating the faulted coil within the stator winding. Fault location would not only expedite repairs but also allow for more optimal repair outage scheduling. This work successfully implemented a theory to predict changes in axial leakage flux resulting from stator winding interturn shorts and developed an algorithm to locate the faulted coil position. An experimental setup comprising a 200 hp motor loaded by a generator validated this theory. Suitable transducers were developed and installed on this motor. Measurements using this experimental configuration clearly confirmed the theoretical model. Based on this experimental work, an instrument for continuous monitoring of shorted turns is under development. According to paper[9], a method was developed to monitor defects such as cracked rotor bars and shorted stator coils in induction motors. Air-gap torque can be calculated during motor operation, eliminating the need for special measurement downtime. Data on air-gap torque should be periodically maintained for comparison. Since this method collects more than just line current data, it offers additional possibilities beyond examining only line current. The theoretical foundation for this proposed method is presented. Experiments on a 5-hp motor demonstrate the validity and potential of this approach. Further studies are planned to expand the proposed method in detail and to monitor defects in other rotating machine types. The health monitoring system proposed in paper[10] is an NI-Compact RIO-based online stator winding fault diagnosis technique for identifying and assessing severity of partial degradation and complete insulation breakdown in stator winding turns of 3-phase induction motors. Principal Component Analysis (PCA) was applied to 3-phase current signals to extract significant features. The variances of the first two principal components (PC1 & PC2) proved critical for fault detection. Inter-turn complete insulation breakdown was distinguished from inter-turn partial insulation failure by analyzing PC2 variance range. Additionally, both fault types' severity was determined by observing PC1 variance trends. Faults involving as little as 0.27% of total phase winding turns could be successfully detected online within milliseconds with acceptable prediction accuracy using k-Nearest Neighbors classifier. Paper[11] proposed a health monitoring system utilizing Remote drive inverters, which represent a new area for enhancing motor efficiency and reliability. With rapid technological advancement and internet network expansion, the Internet of Things has become a primary means of communicating with various devices. The number of internet-connected devices has increased, particularly in industrial environments, leading to improved product control and monitoring. Transmitting system monitoring information via IoT enables processing, storage, and monitoring. IoT application in electric motor control and monitoring has led to smart drive development. IoT benefits include reduced maintenance costs, increased reliability, optimized motor performance, and improved failure prediction accuracy. This project employed a Raspberry Pi board as processor, two

current and voltage sensors for monitoring electrical features and controlling motor speed, and a temperature sensor for detecting failures. The Raspberry Pi collected data and sent it to the server for database storage and processing, after which necessary instructions were returned to the Raspberry Pi.

An online condition monitoring system (OCMS) for transformers is proposed in paper[12], which replaces reactive and preventive maintenance with predictive maintenance. The proposed OCMS offers a cost-effective, online, accurate tool with features like suggesting corrective actions for power utilities' benefit. System results were evaluated by comparison with utility model results. The proposed OCMS applies only to transformers without abnormal faults such as those detected by dissolved gas analysis, distinguishing it from other market-available health index algorithms for transformer condition monitoring, including for abnormal faults. Paper[13] discussed a remote condition monitoring system for distribution transformers. Different parameters were acquired and processed in the remote terminal unit, which communicated data to the operator end via internet. Based on parameter values, the transformer's health index was determined at the operator end interface, with analysis based on this health index. This system differs from power transformer condition monitoring systems in its monitoring techniques and communication methods. A more economical system was designed to precisely evaluate transformer health status. Test results were obtained from a specially designed transformer. The article first explains the system architecture and analysis method, then discusses test results on a distribution transformer. Paper[14] proposed a condition monitoring system using infrared thermal imaging to assess internal faults and detect failures in induction motors. A dataset of 540 infrared thermal images of an induction motor, including five faulty conditions and one healthy condition, was analyzed. After data acquisition, processing, and analysis, histogram and textural features were extracted from the thermal images. Images were classified as hot or cold, and the region of interest was identified using a sliding window technique with auto-thresholding and ground truth. Machine learning classifiers including decision trees, random forest, and ensemble learners like Ada boost and gradient boosting trees were employed to detect and classify faults in both hot and cold conditions. Evaluation metrics assessed each classifier's performance. Paper[15] designed a multi-agent system (MAS) for transformer condition monitoring that utilizes data generated by ultra-high frequency (UHF) monitoring of partial discharge activity. It explains the rationale for using multi-agent techniques and the problems resolved through this technology. Every aspect of the MAS design is discussed, along with the design and performance of intelligent interpretation techniques. Paper[16] designed an IoT-based health monitoring system operating via worldwide networking. This real-time health monitoring system employs temperature sensors, potential transformers, and current transformers to monitor distribution transformer temperature, voltage, and current, transmitting this information to a remote server for monitoring and necessary action to prevent electricity supply outages. The system was laboratory-implemented and tested, with collected data validated and the system confirmed to function satisfactorily. Paper[17] proposed a feedforward multilayer-perceptron Neural Network (NN) trained by back propagation, based on monitoring negative sequence voltage and three-phase shift to check unbalanced supply voltage, which inevitably affects stator currents and therefore the three-phase shift. Data required for NN training and testing were generated using a simulated stator model. Experimental results verified the superior accuracy of the proposed method.

Paper[18] suggests that a method based on reference frame theory can successfully be applied to real-time fault diagnosis of electric machinery systems as a powerful tool for determining fault signature magnitude and phase quantities with good precision. The basic concept involves converting the associated fault signature to a dc quantity, followed by computing the signal's average in the fault reference frame to filter out remaining signal harmonics (ac components). Consequently, neither notch filters nor low-pass filters are needed to eliminate fundamental components or noise. Since incipient fault mechanisms have been extensively studied, motor fault signature frequencies and fault models are well-established. Therefore, ignoring all other components, the proposed method focuses exclusively on certain fault signatures in the current spectrum depending on the examined motor fault. Broken rotor bar and eccentricity faults underwent experimental online testing using a TMS320F2812 digital signal processor (DSP) to demonstrate the method's effectiveness. This application utilized only readily available drive hardware without additional components like analog filters, signal conditioning boards, or external sensors. The DSP served as the motor drive processing unit for both motor control and fault detection, providing instantaneous fault information. The proposed algorithm processes measured data in real time to avoid buffering and large memory requirements, enhancing the method's practicality. Due to the algorithm's short-time convergence capability, fault status updates occur every second. The algorithm's immunity against non-ideal conditions such as measurement offset errors and phase unbalance was theoretically and experimentally verified. As a model-independent fault analyzer, this method applies to all multiphase and single-phase motors.

Paper[19] analyzes the use of motor current signature analysis and mathematical morphology to detect broken bars in induction motors under varying mechanical load conditions. The proposed algorithm first identifies motor load, then motor condition. Statistical analysis of numerous tests under different motor loads (100%, 75%, 50%, and 25%) and motor conditions (healthy, one broken bar, and two broken bars) is presented. The proposed method was implemented in a field programmable gate array for real-time online applications, achieving an average 95% fault detection accuracy. Paper[20] proposes a novel methodology for induction motor BRB detection and fault severity classification using homogeneity as an index—reportedly never before used as a fault diagnosis indicator—analyzing one phase of the induction motor startup-transient current. Due to homogeneity calculation's low computational complexity, this paper proposed a hardware-processing unit based on a field programmable gate array device for online BRB detection and classification. Results demonstrated the high efficiency of the proposed methodology as a deterministic technique for incipient BRB diagnosis in induction motors, capable of detecting and differentiating among half, one, or two BRBs with over 99.7% certainty.

From this study, it's evident that traditional induction motor health monitoring methods employ manual labor and invasive procedures that consume excessive time. In contrast, modern approaches like IoT or Remote Health Monitoring improve time efficiency and are non-invasive but suffer from power consumption issues, critical installation requirements, and potential hardware compatibility problems, including memory limitations or processing hardware compatibility. Remote monitoring systems require reliable internet connectivity to avoid system failure, but most fail to fully utilize modern electronics' computational power or modern processing systems.

The neural network approach in paper[17] employs modern computing methods and algorithms but lacks implementable hardware efficiency in terms of power, time, and area if hardware were developed. FPGA-based approaches offer superior time and power efficiency, though their success rates require further research to develop a health monitoring system that combines time efficiency, power efficiency, and area-efficient implementation.

Paper[21] emphasizes the critical importance of monitoring systems for rotating machines, stressing the need to detect failures at an early stage. Squirrel-cage induction motors, widely used in industry and accounting for 85% of industrial power consumption, face significant challenges from broken rotor bars—a major failure that increases power consumption and can cause additional damage. Previous research primarily focused on current or vibration monitoring, limited to detecting up to one broken bar under specific motor conditions. This paper contributes an innovative methodology for half-broken-bar detection in induction motors, combining current and vibration analysis and correlating signal spectra to enhance detectability in both loaded and unloaded motor conditions. The methodology addresses limitations of isolated techniques struggling to detect such conditions. Implemented on a low-cost field-programmable gate array (FPGA), the paper presents a special-purpose system-on-a-chip (SoC) solution for online operation, including a complex postprocessing decision-making unit. The paper demonstrates the implementation's performance through several case studies.

The success of previous research on methodologies for real-time health monitoring systems has prompted further exploration to develop more robust induction motor health monitoring systems. For instance, Paper [18] proposed a methodology using the TMS320F2812 digital signal processor (DSP) to compute fault signature magnitude and phase quantities, enabling analysis to detect broken rotor bars in induction motors. This approach leveraged DSP technology to accurately process and analyze signal data, enhancing fault detection capabilities. Similarly, Paper [19] introduced an FPGA-based system for detecting broken rotor bars in induction motors, analyzing the motor's current signature combined with mathematical morphology principles to identify faults. The method was extensively tested under various motor operating conditions and loads, demonstrating high fault detection success rates and highlighting the potential of FPGA-based implementations for reliable, efficient industrial monitoring solutions. Additionally, Paper [20] presented another FPGA-based method for detecting broken bar faults in induction motors, focusing on homogeneity estimation through startup transient current analysis, offering a complementary fault detection technique. This method's success underscores FPGA-based systems' effectiveness in detecting faults and ensuring induction motors' operational integrity in industrial settings. Furthermore, Paper [21] proposed an SoC-based system for early detection of half-broken-bar faults in induction motors by correlating vibration and current signal spectra, providing insights into incipient-stage faults and enabling proactive maintenance strategies. SoC technology integration offers advantages in system integration and efficiency, facilitating almost real-time fault detection capabilities. Collectively, these successful research efforts highlight FPGA-based systems' potential for developing advanced induction motor health monitoring solutions. By leveraging innovative technologies and methodologies, these systems promise enhanced reliability, efficiency, and safety in industrial operations through early fault detection and predictive maintenance strategies—a crucial requirement in today's industry and society.

This study explores previous research in the field to develop an enhanced condition monitoring method. Through this investigation, an SoC-based fault detection technique is developed for induction motors to identify possible faults in advance. The SoC receives three-phase stator current data from the motor through a DAS and analyzes it to detect abnormalities in online mode using Park's Vector Approach. The efficiency of this proposed method has been compared with software solutions.

BACKGROUND

This research examines a predictive system for detecting fatigue or faults in induction motors. Such detection requires understanding both normal operating parameters and potential deviations that signal problems. The chapter introduces induction motor fundamentals, operation principles, and fatigue conditions.

Induction Motor

Induction motors are industrial workhorses that power numerous applications across manufacturing, transportation, and utilities. Their popularity stems from their simplicity, durability, and efficiency, making them essential components in modern industry where they reliably drive equipment from conveyor belts to pumps and fans.

Construction and Working Principle of Single-Phase Induction Motor

An induction motor operates through electromagnetic induction between its stator and rotor components:

Stator: The stationary part containing primary and auxiliary windings. The main winding produces a magnetic field when AC voltage is applied, while the auxiliary winding creates a phase shift, generating a rotating magnetic field. When connected to single-phase AC supply, the stator creates a magnetic field that drives the motor at slightly below synchronous speed.

Rotor: The rotating component featuring a squirrel-cage design with conductive bars embedded around its periphery. These bars are short-circuited by end rings at both ends. The rotating magnetic field from the stator induces currents in these bars, creating a magnetic field that interacts with the stator's field to produce rotation. This cage-like structure gives the motor its name. The permanently shorted bars create low electrical resistance, and the absence of slip rings and brushes results in simple, robust construction.

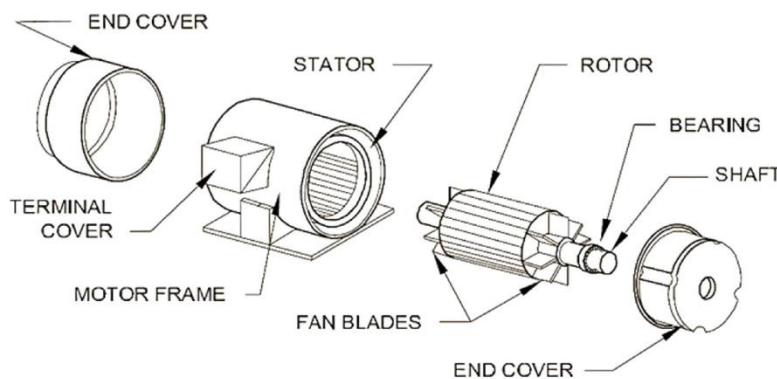


Fig.: Exploded view of Single-Phase Induction motor[22]

The single-phase induction motor differs from polyphase motors mainly by having only one stator winding. When connected to AC power, current flows through this winding, generating flux that links with and cuts across the rotor conductors.

Working Principle

According to Faraday's Law of electromagnetic induction, EMF is induced in the rotor, causing current to flow when the rotor circuit is closed. This generates rotor flux, which gives the induction motor its name. The interaction between the main flux and rotor flux produces the torque needed for rotation.

In single-phase induction motors, the main winding receives single-phase AC current, creating a fluctuating magnetic flux around the rotor. As AC current direction changes, so does the magnetic field direction, but this alone cannot cause rotation.

This is where double revolving field theory applies. According to this theory, a single alternating field can be viewed as a combination of two fields with equal magnitude (half of the alternating field) rotating in opposite directions. When AC is applied, these two half-magnitude fields revolve in opposite directions around the rotor.

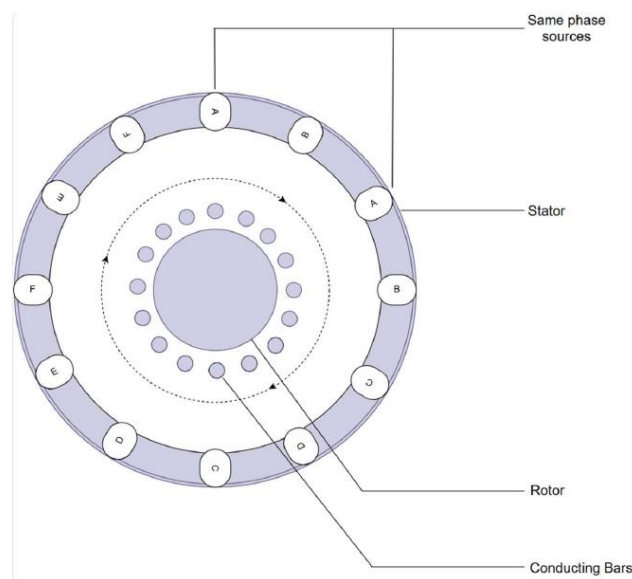


Fig.: Poly-phase Induction Motor basic structure

With current flowing in the stator and magnetic fields revolving around the rotor, Faraday's law takes effect. The revolving magnetic fields induce electricity in the rotor, generating force that rotates the rotor.

Construction and Working Principle of Three-Phase Induction Motor

Here's a rephrased and shortened version:

Three-Phase Induction Motors: Construction and Working Principle

Three-phase induction motors are valued in industrial and commercial settings for their simplicity, efficiency, and self-starting capabilities. They operate through electromagnetic induction, with three-phase currents in the stator creating a rotating magnetic field that induces currents in the rotor, generating the necessary torque for rotation.

Construction Components:

Stator: Contains three-phase windings arranged to produce a rotating magnetic field when connected to three-phase power. This field is key to the motor's operation, inducing currents in the rotor and enabling continuous operation and variable speed control.

Rotor: The rotating component, available in two main types:

- **Squirrel-Cage Rotor:** Most common type featuring aluminum or copper bars embedded in slots around a laminated core and short-circuited by end rings. This design requires no external connections or brushes, offering reliability and minimal maintenance.

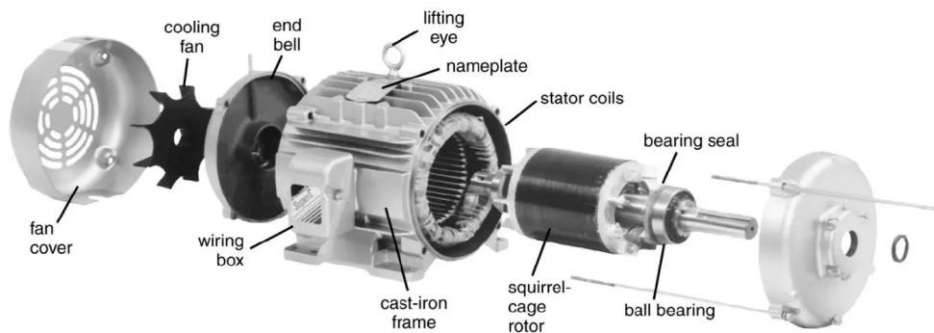


Fig.: Exploded view of Three-Phase Induction Motor's internal construction[23]

- **Wound Rotor:** Features three-phase windings connected to slip rings, allowing external resistors to be connected for speed control and high starting torque. Typically used in applications like cranes, hoists, and elevators.

Bearings: Support the rotor shaft and ensure smooth rotation. These can be ball bearings or sleeve bearings, selected based on motor size and application requirements.

Working Principle of Three-Phase Induction Motors:

When three-phase AC voltage is applied to the stator windings, it generates a rotating magnetic field due to the phase differences. This field induces currents in the rotor conductors through electromagnetic induction. The induced rotor currents create their own magnetic field that interacts with the stator's rotating field, producing torque that causes the rotor to rotate.

The motor's speed is determined by the synchronous speed (dependent on AC frequency and pole count), but the actual running speed is slightly lower due to slip, which is necessary for torque generation. This process effectively converts electrical energy into mechanical energy.

Induction motors serve as workhorses across various industries, powering fans, pumps, compressors, conveyors, lifts, and HVAC systems in both residential and commercial settings. They're also crucial in power generation facilities. Despite their reliability, these motors can develop faults (referred to as Fatigue Conditions) that can cause significant disruptions and costly downtime.

Fatigue in Induction Motors: Causes and Effects

Fatigue in induction motors refers to abnormal behaviors or fault conditions that deviate from normal operation. Several factors can cause these faults:

▪ Main Causes of Induction Motor Faults:

1. Bearing Wear

- Friction from continuous shaft rotation
- Contamination by dust, dirt, and moisture
- Overloading and excessive stress
- Improper lubrication

Excessive wear leads to vibration, noise, and potential complete motor failure.

2. Thermal Failure

- High ambient temperatures
- Operation beyond rated capacity
- Poor ventilation or blocked airflow
- Frequent start-stop cycles

- Voltage fluctuations
- Mechanical misalignment
- Inadequate cooling systems

Overheating degrades insulation, reduces efficiency, and can cause winding failure.

3. **Electrical Faults**

- Winding insulation breakdown
- Rotor/stator winding problems
- Voltage imbalance
- Harmonic distortion

These issues cause unbalanced currents, overheating, and winding damage.

4. **Rotor Faults**

- Broken rotor bars
- Cracks in rotor bars
- Rotor imbalance
- Rotor eccentricity
- End ring faults

These faults require specialized diagnostic techniques and may necessitate component replacement.

5. **Mechanical Misalignment**

- Bearing failures
- Shaft misalignment
- Damage to rotor bars or end rings
- Cooling fan failure

Proper installation and regular maintenance can prevent these issues.

6. **Environmental Causes**

- Temperature extremes
- Humidity and moisture exposure
- Dust and contaminant accumulation

▪ **Effects of Induction Motor Faults:**

The previous study emphasizes the critical importance of maintaining induction motors, which power essential industrial processes, transportation systems, and utility infrastructure. Motor failures can have serious consequences across multiple domains:

Key Impacts of Induction Motor Faults

1. Production Disruption

- Unplanned downtime halts production lines and manufacturing processes
- Decreased output from machinery and equipment impacts overall productivity

2. Financial Loss

- Revenue losses from halted production
- Additional expenses for repairs, replacement parts, labor, and maintenance services

3. Safety Hazards

- Increased risk of accidents and injuries to personnel working near faulty equipment
- Electrical faults may cause overheating, sparking, and potential fires

4. Process Instability

- Product defects or variations compromising quality control standards
- Inconsistent performance affecting production schedules and supply chain logistics

5. Regulatory Compliance Issues

- Potential non-compliance with industry regulations and safety codes
- Risk of legal liabilities, penalties, and damage to organizational reputation

6. Environmental Impact

- Increased energy consumption from inefficient operation
- Higher emissions contributing to environmental degradation

Given these impacts, proactive maintenance strategies are essential to detect potential faults before they escalate into major failures. This creates an opportunity for System on Chip (SoC) technology—integrated circuits combining multiple components like processing units, memory, peripherals, and interfaces on a single chip—to enable more efficient, powerful, and secure fault detection systems for induction motors.

SoC Architecture

The SoC (System on Chip) architecture combines Programmable Logic (PL) and Processing System (PS) components to create versatile, powerful, and secure processing systems.

Key Components

1. Programmable Logic (PL)

- Flexible logic fabric similar to FPGAs
- Enables implementation of custom digital circuits
- Includes DSP blocks, look-up tables, flip-flops, and configurable logic elements

2. Processing System (PS)

- Contains main processing units (CPUs)
- Includes memory controllers and various peripherals
- Executes software applications and manages system functions
- Configurable via software rather than hardware programming
- Can function as data processor, graphical processor, or arithmetic/logic processor

3. PS-PL Integration Benefits

- Forms a heterogeneous computing architecture
- PS handles general-purpose processing and operating systems
- PL manages specific high-performance digital logic tasks
- Enables custom accelerators and optimized solutions
- Provides balance between flexibility and performance

4. Memory Integration

- PS integrates with memory subsystem
- Enables efficient data access and management
- Crucial for smooth software execution

5. Data Security

- Strong encryption protocols between PS-PL interconnections
- Integrated memory provides protection against data corruption
- Secure intercommunication and processing
- Uses security encryption protocols for multiple design blocks

This architecture creates a platform where software tasks are handled by the PS while hardware acceleration and custom logic are implemented in the PL, delivering both flexibility and performance.

High Level Synthesis and System Design

C-based VLSI (Very Large-Scale Integration) design refers to the use of high-level programming languages, primarily C or C-like languages, for hardware design and modeling.

This approach is a significant departure from traditional methods, which rely on Hardware

Description Languages (HDLs)

like Verilog or VHDL for circuit-level specification. By leveraging the familiarity and abstraction of C, engineers can design complex hardware systems at a higher level of abstraction, focusing on functionality rather than detailed hardware intricacies.

In C-based VLSI design, High-

Level Synthesis (HLS) tools play a pivotal role by converting C or C++ descriptions into RTL (Register Transfer Level) code, which can then be synthesized into physical hardware.

This approach allows for faster design iterations, improved productivity, and easier debugging compared to HDL-based design. It is particularly valuable in developing large-scale integrated circuits such as processors, FPGAs, and SoCs (System-on-Chip), where reducing time-to-market is critical.

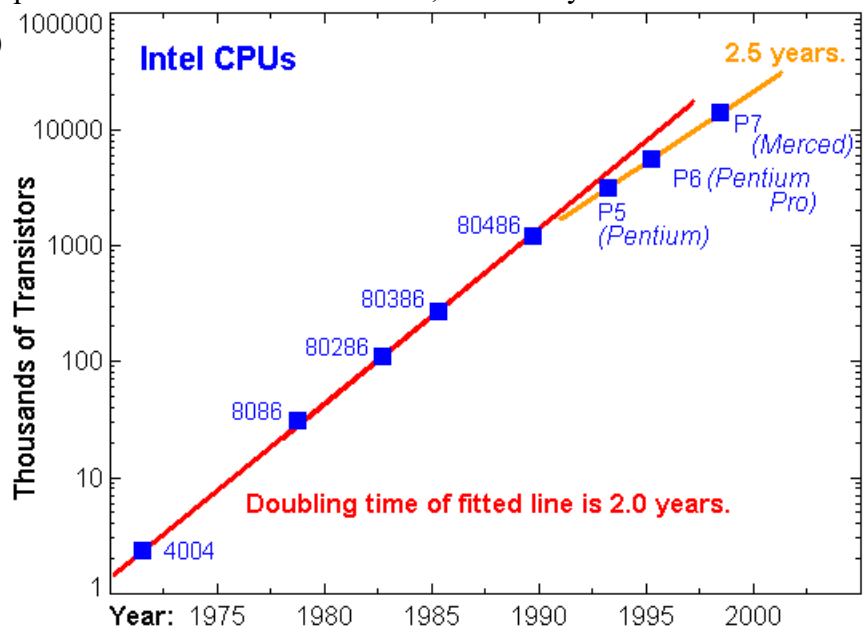


Fig.: design trends over the year

THE VLSI DESIGN FLOW

The steps are as follows:

- 1. System Specifications:** The initial step where the overall functionality, performance requirements, power consumption, area, and cost of the system are defined. This stage involves identifying the key features of the system, such as processing speed, data bandwidth, and power efficiency.
- 2. Architectural Design:** The high-level design of the system is created based on the specifications. This includes defining the system's data paths, control paths, functional units (e.g., ALUs, memory units), communication interfaces, and overall structure (e.g., pipelining, parallelism).
- 3. High-Level Synthesis (HLS):**
High-Level Synthesis is the process of automatically generating RTL code from high-level descriptions (such as C/C++ or SystemC). In this stage, the design described in a high-level language is mapped into an RTL representation that describes how data moves between registers and is processed using combinational logic.
- 4. Logic Synthesis:** Logic synthesis transforms the RTL description into a gate-level netlist. This process maps the functional description of the system into basic logic gates (AND, OR, NOT, etc.), using standard cells from a technology library.
- 5. Physical Design:** After physical design, the layout is converted into masks, which are used to fabricate the chip on silicon wafers using semiconductor manufacturing techniques such as photo-lithography, doping, and etching.
- 6. Fabrication:** After physical design, the layout is converted into masks, which are used to fabricate the chip on silicon wafers using semiconductor manufacturing techniques such as photo-lithography, doping, and etching.
- 7. Packaging & Testing:** Once the chip is fabricated, it is packaged to protect it from physical damage and environmental factors while providing the necessary electrical connections. After packaging, the chip is tested for functionality, performance, and reliability.

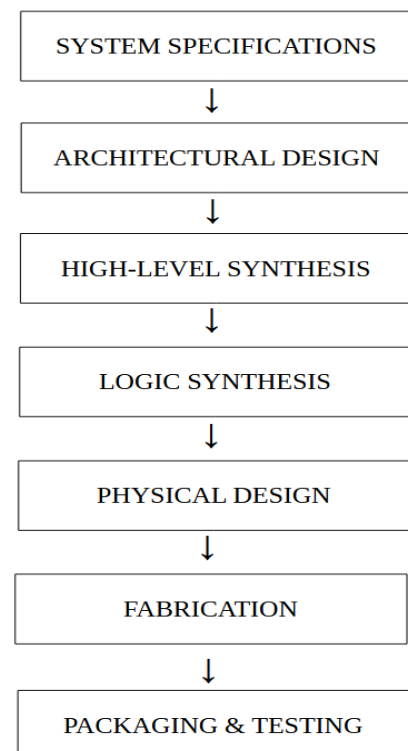


Fig.: The VLSI design flow

THE HLS DESIGN FLOW

The High-Level Synthesis steps are as follows:

1. **Preprocessing:** Intermediate representation, checking for data-dependency, live variable analysis, control and data flow graph. The tasks for different timestamps are assigned and the inter-dependency of the variables are analyzed. It reduces the complexity of the code, making it easier to translate into hardware without errors or inefficiencies.
2. **Scheduling:** Assigning control step to the operations of the input behavior. Assigning the operations in the subsequent timestamps. The tasks are assigned and Scheduling ensures that each operation in the high-level description (e.g., C/C++) is assigned a specific clock cycle for execution.
3. **Register Allocation:** Computing minimum number of functional units and registers, maximizing resource sharing. During the execution of a program or hardware design, intermediate values and final results need to be stored temporarily. These variables are allocated to registers, which are fast memory elements capable of storing values in hardware. In hardware, the number of available registers is limited. Register allocation aims to minimize the number of registers used, especially when resources are scarce.
4. **Binding:** Variables are mapped to registers, operations to functional units, data transfers to the interconnection units. Operation binding involves mapping high-level operations (such as arithmetic, logical, or control operations) to specific hardware functional units. For example, an addition operation in the high-level code needs to be bound to a specific adder in the hardware.
5. **Data path & Controller Design:** Control unit is designed based on inter connections among the data path elements, data transfer required in different control steps. Together, the data-path and controller work in harmony to execute a given task, forming the core of digital hardware designs. These components together form the backbone of most digital systems such as processors, custom accelerators, and other hardware designs.

```

while (x_var < a_var) loop
  t1 := u_var * dx_var;
  t2 := 3 * x_var;
  t3 := 3 * y_var;
  t4 := t1 * t2;
  t5 := dx_var * t3;
  t6 := u_var - t4;
  u_var := t6 - t5;
  y1 := u_var * dx_var;
  y_var := y_var + y1;
  x_var := x_var + dx_var;
end loop;
X <= x_var;
Y <= y_var;
U <= u_var;
  
```

Fig.:Preprocessing

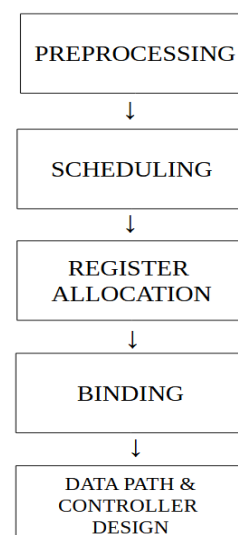
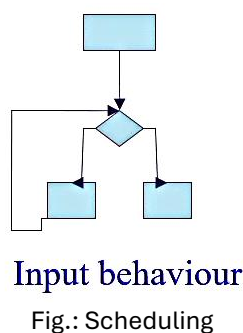


Fig.: HLS design flow

R1 : 3, v1
 R2 : x u, v5
 R3 : v0, v6
 R4 : v3

 FU1: op1, on3. ...
 FU2: op2, op5, ...
 FU3: ...

Fig.: Allocation & Binding

TYPICAL C/C++ CONSTRUCTS TO RTL

Below are some of the most commonly used C/C++ constructs which are frequently needed to be converted to their respective RTL counterparts. The mappings are shown below:

<i>C/C++ constructs</i>		<i>Hardware components</i>
<i>Functions</i>	→	<i>Modules</i>
<i>Arguments</i>	→	<i>Input/Output ports</i>
<i>Operators</i>	→	<i>Functional units</i>
<i>Scalars</i>	→	<i>Wires or Registers</i>
<i>Arrays</i>	→	<i>Memory blocks</i>
<i>Control flows</i>	→	<i>Control logics</i>

Some of them are discussed in the subsequent sections:

1. Function Hierarchy:

- Each function is translated into a RTL module. The top function becomes the top module and so on.
- The hierarchy of the modules are maintained accordingly.
- For example, we have functions called A and B, so it will create module A which will implement this function A and it will create another module B.

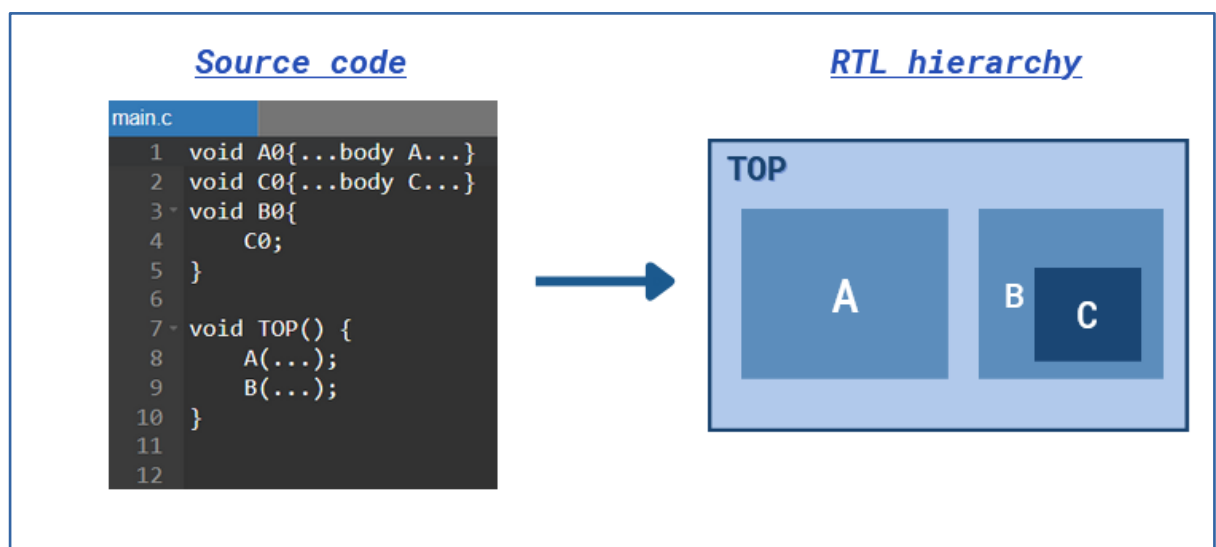


Fig.: Function Hierarchy

2. Arrays:

- By default, an array in C code is typically implemented by a memory block in the RTL.
- An array can be partitioned and mapped to multiple RAMs.
- A large array can also be partitioned into individual elements and mapped to registers.
- Multiple small arrays can be merged and mapped into one RAM.

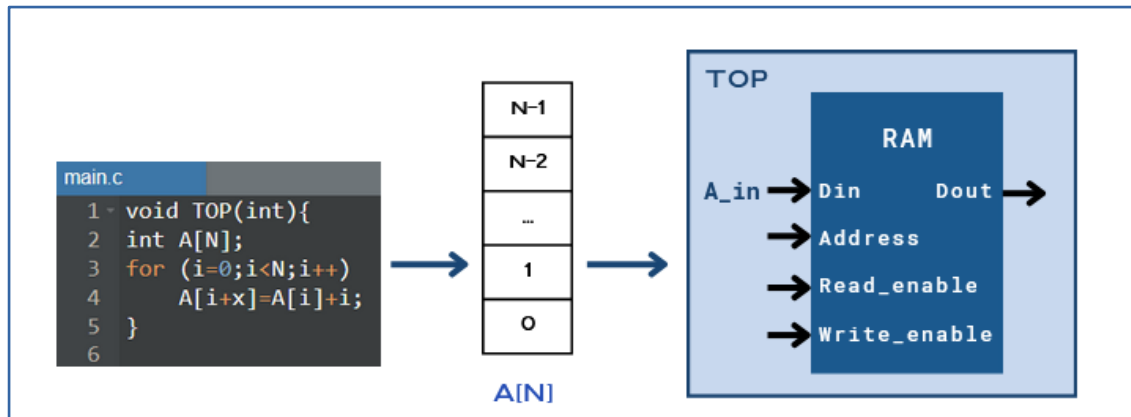


Fig.: Arrays

3. Loops:

- By default, loops are rolled.
- Loops are **unrolled** to achieve shorter latency & increase parallelism.
- Each loop iteration corresponds to a 'sequence' of states.
- But it will in turn increase area & power.
- So, the loops are **pipe-lined** for an optimized output.
- It allows a new iteration to begin processing before the previous iteration is complete.

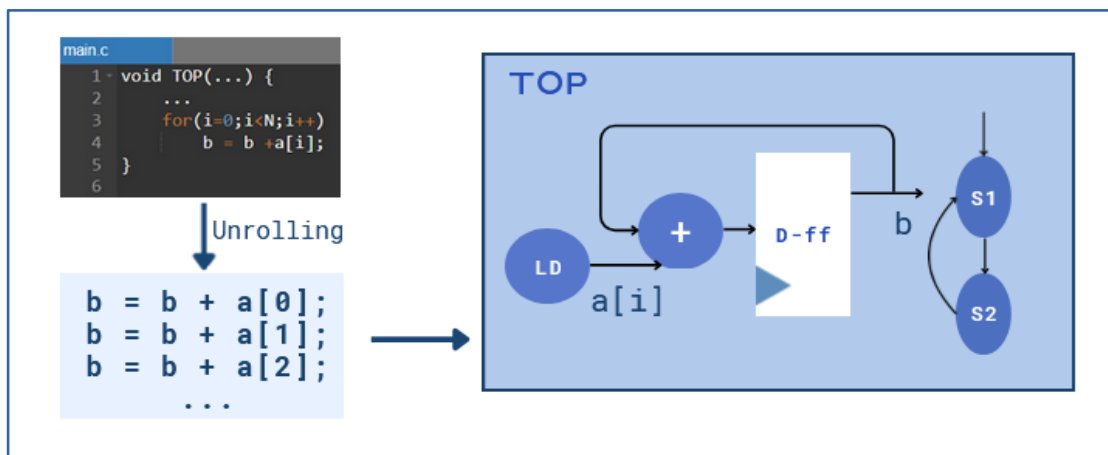


Fig.: Loops

Bitstream Generation and Hardware Synthesis

After synthesizing, placing, and routing a design, bitstream generation translates the final circuit description into a binary format that configures the FPGA device. This process defines the behavior and functionality of programmable logic elements and interconnections.

Key Steps in Bitstream Generation

- **Configuration Data Generation:** Creates binary values that program logic elements and routing resources on the FPGA/PL portion of SoC
- **Compression and Encoding:** Applies techniques like Huffman coding or run-length encoding to reduce bitstream size and improve programming efficiency
- **Checksum and Error Correction:** Adds verification codes to ensure bitstream integrity during configuration
- **Partitioning and Partial Reconfiguration:** Enables updating specific regions without reprogramming the entire device
- **Device-Specific Formatting:** Adds headers, footers, and metadata for compatibility with the target device

Role of High-Level Synthesis and Hardware Synthesis

High-Level Synthesis (HLS) allows designers to express complex algorithms using languages like C, C++, or SystemC, transforming them into hardware implementations optimized for FPGAs or ASICs. This raises the abstraction level, enabling rapid prototyping and efficient implementation.

Hardware Synthesis maps the logical circuit description to concrete locations on the target device, optimizing wire lengths, resource utilization, and timing. The resulting bitstream configures the device's behavior and functionality.

For induction motor fault detection using SoC architecture, these technologies are critical for developing efficient hardware implementations of detection algorithms. HLS expresses sophisticated signal processing algorithms in high-level languages, while Hardware Synthesis translates them into optimized hardware architectures. This enables real-time monitoring and early fault detection, enhancing reliability, efficiency, and safety in industrial applications.

METHODOLOGY

Previously, the focus was on elucidating the phenomenon of fatigue conditions in induction motors and exploring the potential benefits of System-on-Chip (SoC) architecture in early detection of such conditions. Additionally, the utilization of High-Level Synthesis (HLS) for implementing SoC designs, particularly the Hardware Description Language (HDL) required for the design, was discussed. Building upon this theoretical foundation, this chapter delves into the practical aspects of the research by detailing the methodology employed in this work.

Park's Vector

Park's Vector transformation, also known as dq transformation, is a fundamental concept in the field of electrical engineering, particularly in the realm of electric machines and power systems. Developed by the renowned engineer and scientist Robert H. Park, this mathematical technique plays a pivotal role in the analysis and control of alternating current (AC) systems, enabling the representation and manipulation of three-phase quantities in a two-coordinate reference frame. At its core, Park's Vector transformation facilitates the conversion of variables defined in the stationary three-phase reference frame (abc) to a rotating two-coordinate reference frame (dq), where d-axis aligns with the direct component of the system and q-axis represents the quadrature component. This transformation simplifies the analysis and control of AC systems, especially in applications such as motor drives, renewable energy systems, and power electronics.

Mathematical Representation

As stated in previous chapter in the three-phase induction motor's the stator winding three-phase ac voltage is applied. The voltages can be denoted as V_a , V_b , V_c as shown in fig 4.1.1 and they are apart from each other by a phase of 120° or $2\pi/3$ radians.

Mathematically the winding voltages can be described as:

$$\begin{aligned}V_a &= V_m \cos(\omega t + \phi_i) \\V_b &= V_m \cos(\omega t + \phi_i + \frac{2}{3}\pi) \\V_c &= V_m \cos(\omega t + \phi_i + \frac{4}{3}\pi)\end{aligned}$$

where, V_m = Peak voltage delivered by the power supply

ϕ_i = Initial phase of the voltage

ω = angular frequency of the supply voltage

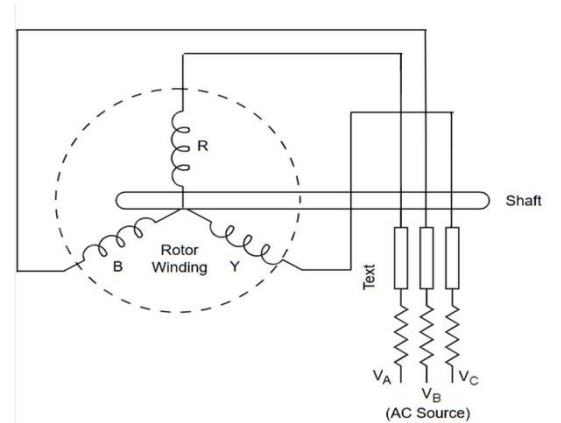


Fig.: Three phase Motor connection with source

Corresponding winding currents are I_a, I_b, I_c and mathematically the currents can be described as :

$$I_a = I_m \cos (\omega t + \varphi_i)$$

$$I_b = I_m \cos (\omega t + \varphi_i + \frac{2}{3}\pi)$$

$$I_c = I_m \cos (\omega t + \varphi_i + \frac{4}{3}\pi)$$

φ_i = Initial phase of the voltage, ; $\theta_i = \phi_i - \text{angle}(Z)$

$\text{angle}(Z)$ = phase of winding Impedance

ω = angular frequency of current

The stator winding current that is in static three phase reference frame is translated into a rotating two-dimensional frame that basically is the rotors reference frame or in simple words how the rotor sees the current and voltage in stator winding. This transformation is known as Park's vector, named after the scientist, mentioned before.

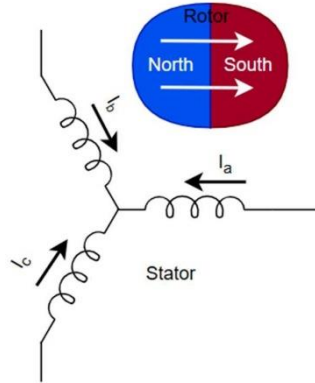


Fig.: Stator current with respect to Rotor

The first step of Park's Vector approach is Clarke Transformation. Clarke Transformation, also known as $\alpha\beta$ Transformation, was devised by Edith Clarke to convert the time-domain components of a three-phase system into two components in an orthogonal stationary frame. As defined previously I_a, I_b, I_c denote the three-phase current flowing through the stator windings.

Let, I_α, I_β be the current flowing along the α and β axis shown in Fig. 4.1.3.

The angles between I_a, I_b, I_c are 120° or $\frac{2}{3}\pi$ radians, that is Angle between I_a and I_b = Angle between I_b and I_c = Angle between I_c and I_a = 120°

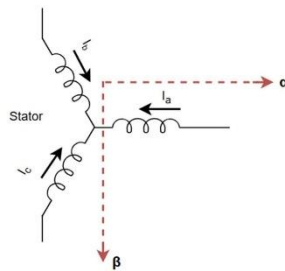


Fig.: Positioning of $\alpha\beta$ axes with respect

As, the three-phase winding is a balanced system so by Kirchhoff's Current Law

$$I_a + I_b + I_c = 0$$

$$I_b = I_c = -\frac{1}{2}I_a$$

So, using trigonometry it can be formulated that

$$I_\alpha = I_a - I_b \cdot \cos\left(\pi - \frac{2}{3}\pi\right) - I_c \cdot \cos\left(\frac{4}{3}\pi - \pi\right) = I_a - \frac{1}{2}I_b - \frac{1}{2}I_c$$

$$I_\beta = I_b \cdot \cos\left(\frac{2}{3}\pi - \frac{1}{2}\pi\right) - I_c \cdot \cos\left(\frac{3}{2}\pi - \frac{4}{3}\pi\right) = \frac{\sqrt{3}}{2}I_b - \frac{\sqrt{3}}{2}I_c$$

So, we can say:

$$I_0 = \frac{1}{2}I_a + \frac{1}{2}I_b + \frac{1}{2}I_c = 0,$$

So, in a matrix format:

$$I_{\alpha\beta} = \begin{bmatrix} I_\alpha \\ I_\beta \\ I_0 \end{bmatrix} = \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \cdot \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix}$$

So, if $I_a = -1$ unit then $I_b = I_c = 1/2$ using, that in equation

$$\begin{bmatrix} I_\alpha \\ I_\beta \\ I_0 \end{bmatrix} = \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \cdot \begin{bmatrix} -1 \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

So,

That leaves us with, $I_\alpha = -\frac{3}{2}$ or in more general terms $I_\alpha = -\frac{3}{2}I_a$

But we need to calculate the actual current in each axis, so $I_\alpha = -I_a$

For balancing the equations in magnitude

$$\begin{bmatrix} I_\alpha \\ I_\beta \\ I_0 \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \cdot \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix}$$

So, $I_{\alpha\beta}$ can be defined in terms of I_{abc}

From this $I_{\alpha\beta}$, dq-transformation can be derived easily by taking account of the rotor magnetic field and rotor's angular speed; more specifically rotor's angular position with respect to the $\alpha\beta$ frame or $\alpha\beta$ axes.

In dq-transformation d and q denotes two axes of dq reference frame and the specific orientation of the two axes. 'd' denotes the axis that is aligned with the magnetic field of the rotor and called the 'Direct Axis' and 'q' denotes 'Quadrature Axis' which is aligned with a direction 90° counterclockwise to rotor's magnetic field(north to south) as shown in Fig. 4.1.4.

The physical significance of dq reference frame is when the current in the 'd' axis increases so does the torque, so the 'd' contributes to torque on rotor and the current in the 'q' axis generates a magnetic field of its own which contribute in total magnetic field by adding to or subtracting from rotor's magnetic field.

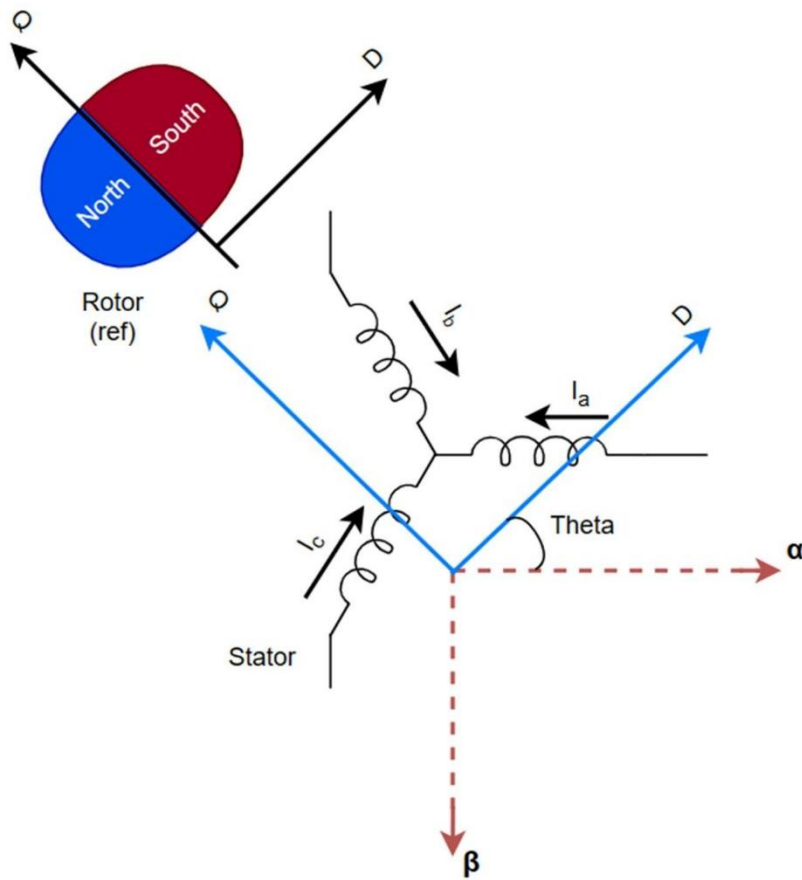


Fig.: dq-reference frame with respect to rotor and stator and αβ axes

So, if the rotor has an angular speed of ω_r radian per second and after t seconds achieved the position shown in the above figure.

So, the angle in between dq-axes and αβ-axes is $\omega_r t$

So, angle between D and α = angle between Q and β = $\theta = \omega_r t$

So, by applying trigonometric ratios

$$I_d = I_\alpha \cos \theta + I_\beta \sin \theta = I_\alpha \cos(\omega_r t) + I_\beta \sin(\omega_r t)$$

$$I_q = -I_\alpha \sin \theta + I_\beta \cos \theta = -I_\alpha \sin(\omega_r t) + I_\beta \cos(\omega_r t)$$

In matrix format we can write and using $I_{\alpha\beta}$ from equation

$$\begin{bmatrix} I_d \\ I_q \end{bmatrix} = \begin{bmatrix} \cos(\omega_r t) & \sin(\omega_r t) \\ -\sin(\omega_r t) & \cos(\omega_r t) \end{bmatrix} \begin{bmatrix} I_\alpha \\ I_\beta \end{bmatrix} \quad (4-1)$$

$$\text{or, } \begin{bmatrix} I_d \\ I_q \end{bmatrix} = \frac{2}{3} \begin{bmatrix} \cos(\omega_r t) & \sin(\omega_r t) \\ -\sin(\omega_r t) & \cos(\omega_r t) \end{bmatrix} \cdot \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix}$$

$$\text{or, } \begin{bmatrix} I_d \\ I_q \end{bmatrix} = \frac{2}{3} \begin{bmatrix} \cos(\omega_r t) & -\frac{1}{2}\cos(\omega_r t) + \frac{\sqrt{3}}{2}\sin(\omega_r t) & -\frac{1}{2}\cos(\omega_r t) - \frac{\sqrt{3}}{2}\sin(\omega_r t) \\ -\sin(\omega_r t) & \frac{1}{2}\sin(\omega_r t) + \frac{\sqrt{3}}{2}\cos(\omega_r t) & \frac{1}{2}\sin(\omega_r t) - \frac{\sqrt{3}}{2}\cos(\omega_r t) \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix}$$

Now taking account for initial condition, that is $t = 0$

$$\begin{bmatrix} I_d \\ I_q \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix}$$

If we take account of I_0 in dq transformation then the matrix can be modified as below

$$\begin{bmatrix} I_d \\ I_q \\ I_0 \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix}$$

There is also a variant of dq-transformation that translates power from static abc-frame to a rotating dq-frame, this variant is also known as the power variant of dq- transformation.

In matrix format the power variant is written as

$$\begin{bmatrix} P_d \\ P_q \\ P_0 \end{bmatrix} = \sqrt{\frac{2}{3}} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} P_a \\ P_b \\ P_c \end{bmatrix}$$

Advantages of Park's Vector Approach

Park's Vector Analysis or dq-transformation offers several advantages, including the decoupling of the direct and quadrature components, which allows for independent control of torque and flux in AC machines. By decoupling these components, it becomes possible to achieve precise control over motor speed, torque, and reactive power, thereby enhancing the performance and efficiency of electric machines.

Furthermore, Park's Vector approach enables the representation of unbalanced and transient phenomena in a stationary reference frame, providing insights into the dynamic behavior of AC systems under varying operating conditions. This capability is particularly valuable in fault detection, transient stability analysis, and control system design, where accurate representation and analysis of system dynamics are essential.

Implementation of Park's Vector through Mathematical Model

From previous discussion we can summarize that Park's Vector Approach provides us with a mathematical model that helps us to translate the three-phase stator current from static reference frame to the dq frame.

Using the dq-transform, another mathematical model is devised that takes the current data as input and analyse the current data to detect any abnormality. The mathematical model is implemented in MATLAB to check its efficiency.

Mathematical Model

The mathematical model leverages the power of Park's vector by recognizing geometrical pattern in the three-phase current data of induction motor with different loading condition and different health condition, any deviation from the pattern derived from a healthy induction motor is pointed out as potential faulty condition

The model first records the three-phase current data and translates the three-phase current data to dq-axes by the matrix previously described. From this conversion the Id and Iq data is stored and Parks vector is created using Id and Iq by the following method

$$Pv = I_d + jI_q$$

The Park's Vector, Pv is analysed to find the data cycle, that is a complete cycle of data or Park's Vector representing the behaviour of the induction motor. The data cycle is found by simply checking when the imaginary component of the Park's Vector changes its sign or crosses zero. Three consecutive zero-crossing points are detected and all the current data of both Id and Iq are stored as the data of one complete cycle.

The data of one cycle is then used to used to determine diagonals by applying Euclidean distance formula. The distance between all the points are calculated using the distance formula and stored as diagonals. Along with the diagonals, using the concept of slope the intercept regarding to each diagonal is calculated which in turn gives us the two end points of the diagonal.

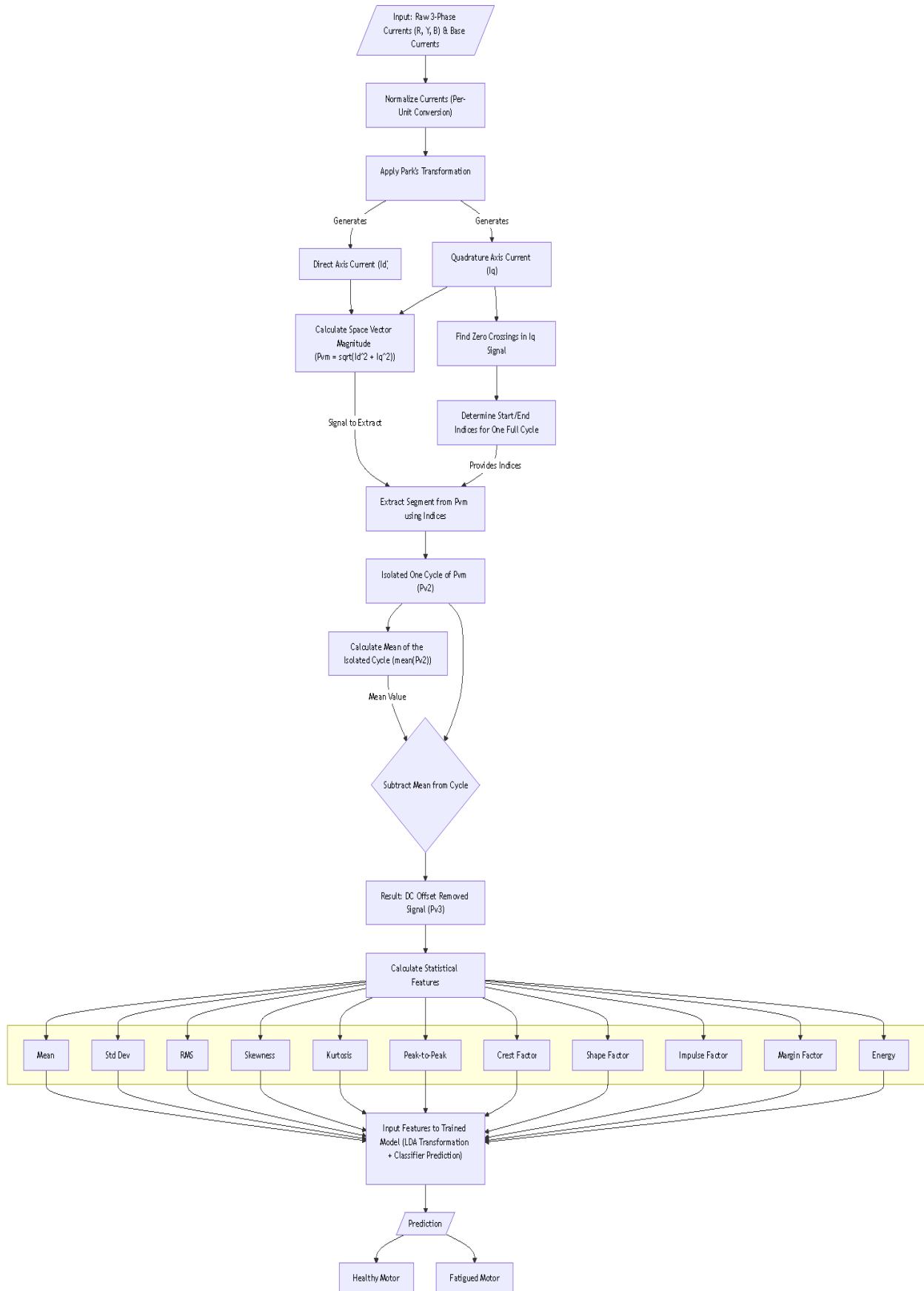


Fig.: Flowchart presentation of the mathematical model

IMPLEMENTATION & RESULTS

Platform Overview: Xilinx PYNQ-Z2

The Xilinx PYNQ-Z2 board is a powerful embedded system development platform that integrates both software and hardware capabilities, enabling efficient hardware-software co-design.

Processing System (PS):

- **Core:** Dual-core ARM Cortex-A9 processor
- **Functionality:**
 - Runs an embedded Linux operating system
 - Supports high-level software development using Python
 - Executes the algorithm purely in **software**

Programmable Logic (PL):

- **Component:** FPGA fabric (Zynq-7000 SoC)
- **Functionality:**
 - Allows the design and deployment of custom hardware logic
 - Utilizes High-Level Synthesis (HLS) to convert C/C++ to RTL
 - Executes the algorithm in **hardware**

Implementation Workflow

The algorithm was implemented in two different environments to facilitate performance benchmarking:

1. Software Implementation:

- **Language:** Python
- **Execution Environment:** Linux OS running on ARM Cortex-A9 (PS)
- **Purpose:** Serve as a baseline for performance measurement using traditional software execution.

2. Hardware Implementation:

- **Method:** High-Level Synthesis (HLS)
- **Deployment Target:** Programmable Logic (PL) of the FPGA
- **Purpose:** Accelerate the algorithm using parallel, pipelined custom hardware logic.

Objective

The primary goal of this experiment is to:

- **Directly compare the execution speed** between:

- **Software implementation** on ARM processors, and
- **Hardware acceleration** using the FPGA fabric.
- Demonstrate the **performance benefits of hardware offloading** and parallelism achievable through FPGA-based design.

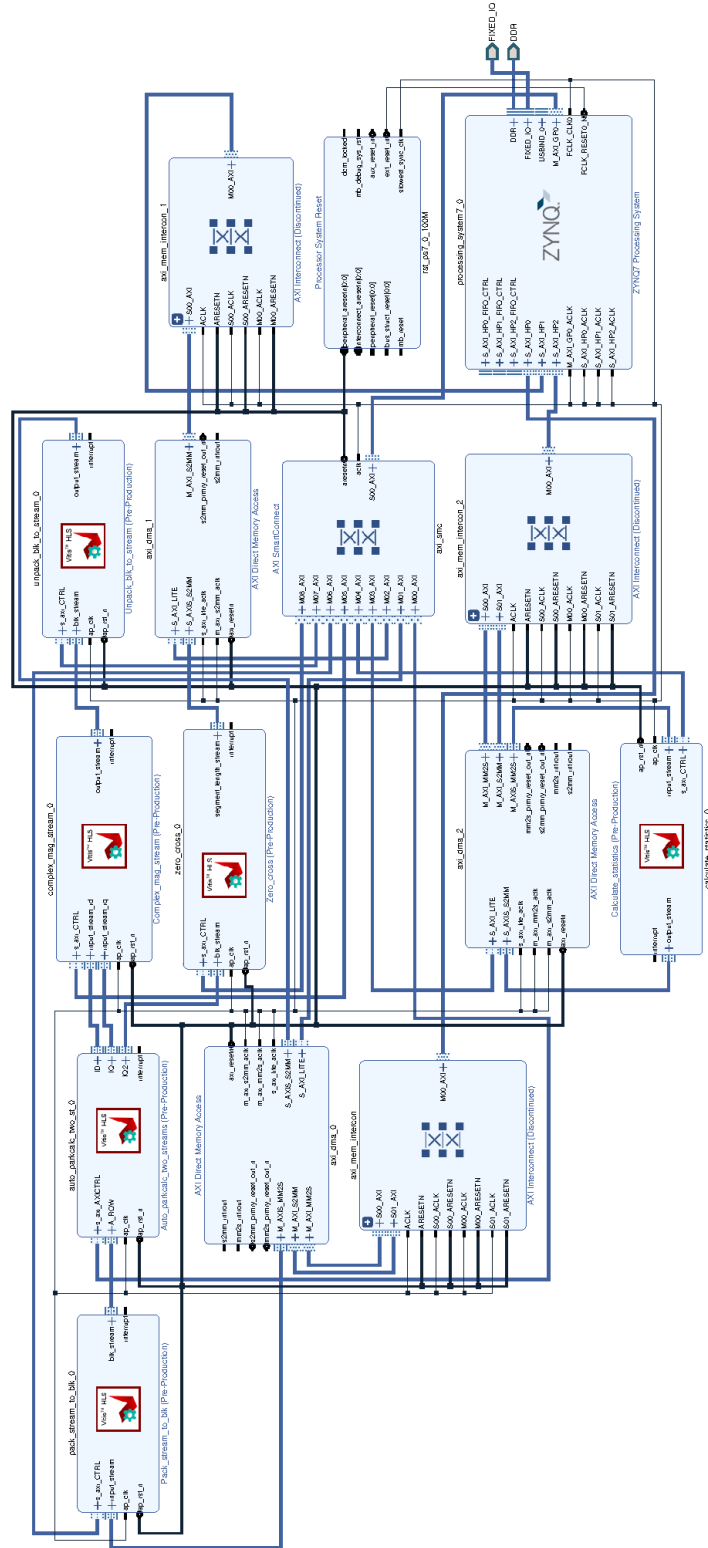


Fig.: Flowchart presentation of the circuit

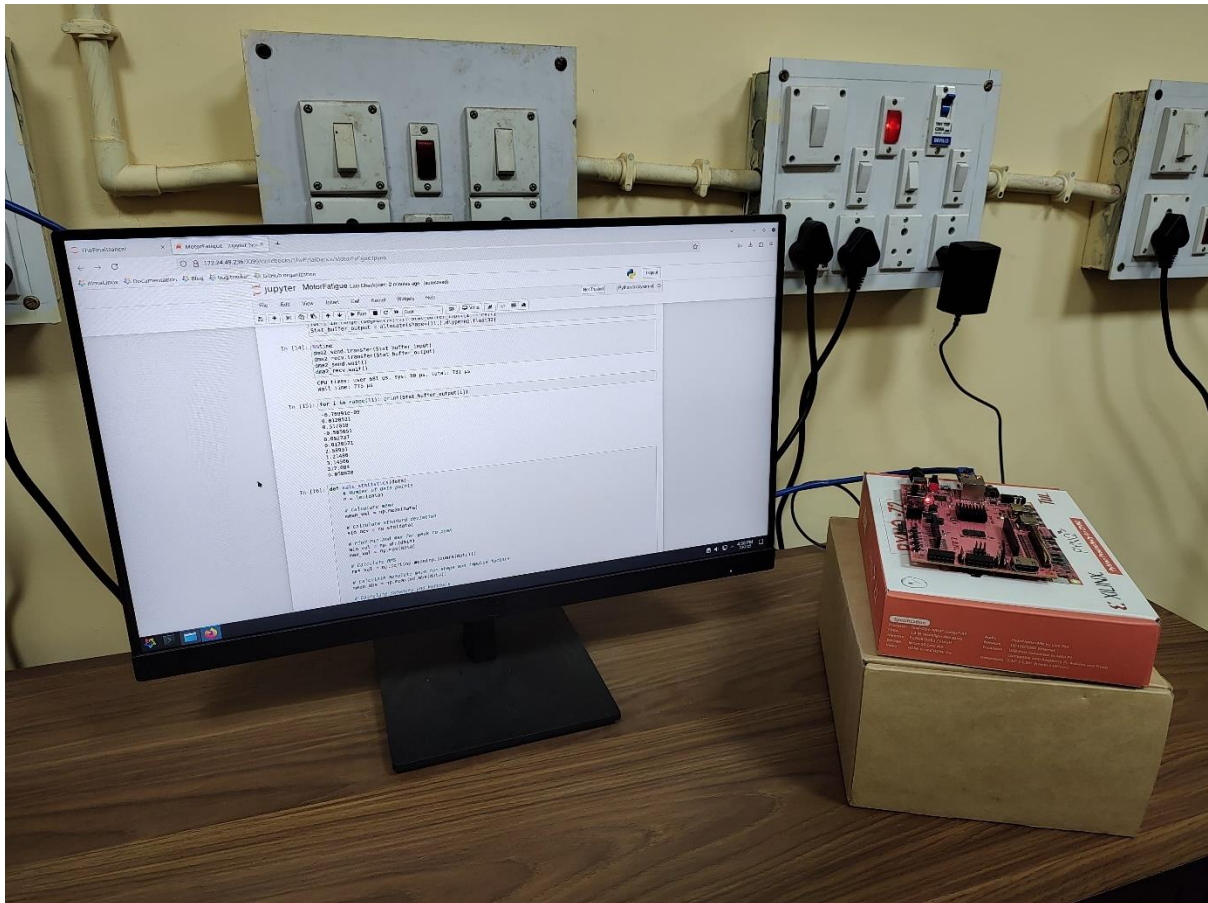


Fig.: Execution

Execution Time Comparison: Software vs. Hardware

1. Software Implementation (Python on ARM Cortex-A9):

- **Platform:** Processing System (PS) of the PYNQ-Z2 board
- **Language/Environment:** Python (running on Linux OS)
- **Execution Time:** 20.46 milliseconds
- **Characteristics:**
 - Sequential execution model
 - Limited performance due to general-purpose nature of ARM processor
 - Higher latency due to software overhead and interpreted language execution

2. Hardware Implementation (FPGA via HLS):

- **Platform:** Programmable Logic (PL) of the PYNQ-Z2 board

- **Technique:** High-Level Synthesis (HLS) to generate custom logic
- **Execution Time: 1.531 milliseconds**
- **Characteristics:**
 - Custom datapaths enable parallel and pipelined execution
 - Exploits hardware-level concurrency
 - Drastically reduced latency due to removal of software stack and optimized computation

Speedup Calculation

$$\text{Speedup} = \frac{\text{Software Time}}{\text{Hardware Time}} = \frac{20.46 \text{ ms}}{1.531 \text{ ms}} \approx \mathbf{13.36\times}$$

Interpretation of Results

- **Hardware acceleration provided a performance boost of over 13×**, significantly reducing the execution time of the feature extraction pipeline.
- This speedup **demonstrates the advantages of FPGA-based computing**, especially for:
 - **Compute-intensive operations** (like feature extraction)
 - **Time-critical embedded applications**
 - **Tasks that benefit from data-level or task-level parallelism**

Gains & Benefits of Hardware Acceleration

The gains and benefits of hardware acceleration are:

- Significant Speedup (~13.36x)
- Lower Latency for rapid detection
- Enables true edge processing:
 - Reduces data transmission requirements
 - Enhances privacy and security
- CPU Offloading: ARM cores can run the final ML model and handle communications
- Potential for improved power efficiency

Use Cases & Applications:

- **Predictive Maintenance:** Schedule maintenance based on detected fatigue
- **Industrial Automation:** Monitor motors in assembly lines, CNC machines, robotic arms
- **HVAC Systems:** Monitor fans, pumps, and compressors in large buildings
- **Renewable Energy:** Monitor generators in wind turbines or solar arrays
- **Smart Edge Devices:** Integrate into smart sensors or embedded monitoring systems
- **Automotive (Non-critical):** Monitor auxiliary motors (pumps, fans) in vehicles.

CONCLUSION

This work marks a significant milestone in condition monitoring of induction-based machines, introducing a pioneering approach built on System-on-Chip (SoC) architecture. By incorporating a novel hardware accelerator, it offers a robust and efficient method for early fault detection in induction motors.

The process begins with evaluating an algorithm based on Park's Vector Approach, followed by its successful hardware validation using High-Level Synthesis (HLS). This leads to the generation of an Intellectual Property (IP) core in HDL, implemented within the Programmable Logic (PL) of the SoC. Hardware prototyping results demonstrate the approach's effectiveness in detecting early signs of motor fatigue, as previously discussed.

The core objective of this study is to leverage SoC technology for real-time health monitoring of motors—enhancing reliability, reducing downtime, and improving safety. Proactive fault detection enables optimized maintenance schedules, cost savings, and extended equipment life. It also promotes energy efficiency and aligns with sustainability goals.

Remote monitoring and predictive capabilities further empower industries with data-driven decision-making and continuous operational improvement. Although this work primarily focuses on prototyping, it lays a solid foundation for future developments, including full- or semi-ASIC implementations for real-world deployment.

Looking ahead, the approach can be extended to other induction-based machines such as generators and transformers. Adapting the algorithm for broader applications could further revolutionize health monitoring across various industrial domains.

In conclusion, this study represents a significant advancement at the intersection of science, engineering, and industry. By harnessing SoC technology for predictive maintenance, it not only addresses key industrial challenges but also paves the way for smart infrastructure and next-generation automation. Through continued innovation and collaboration, this work sets a precedent for the future of proactive, data-driven health monitoring in modern industry.

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